

Making Deep Networks Trainable

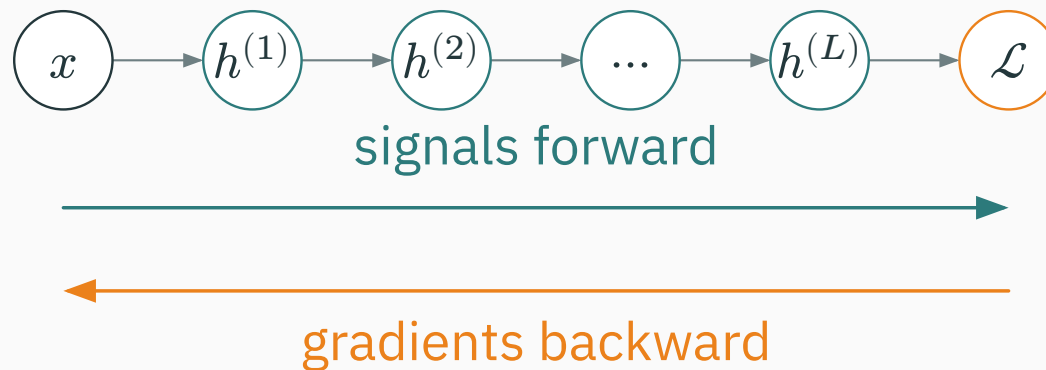
Initialization, activations, normalization and residual connections

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ES 667 · Deep Learning · IIT Gandhinagar

Why depth breaks naive networks

A deep network must preserve useful **signals** forward
and useful **gradients** backward.



everything today keeps these two flows from collapsing or exploding

We have the ingredients — so why stall?

We can already build a net (neurons, nonlinearities), get exact **gradients** (backprop), and **use** them (SGD / Adam). So why not stack 100 layers?

Stack many layers with textbook choices (sigmoid, small random weights) and training **stalls or blows up**. A 100-layer plain net often does **worse** than a 10-layer one — not overfitting, it will not **optimize**.

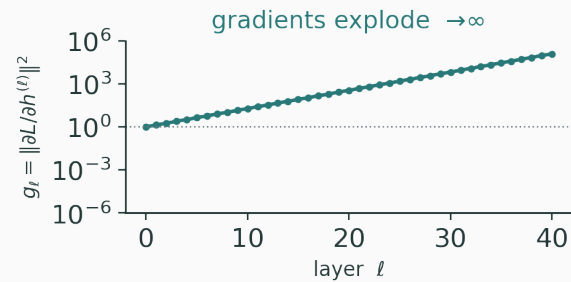
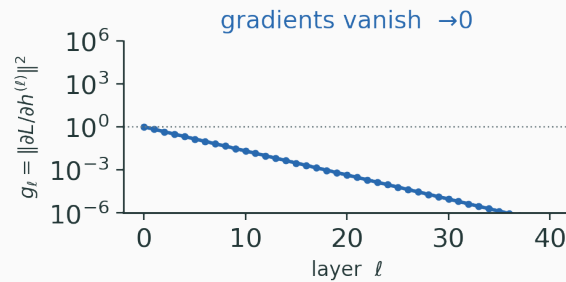
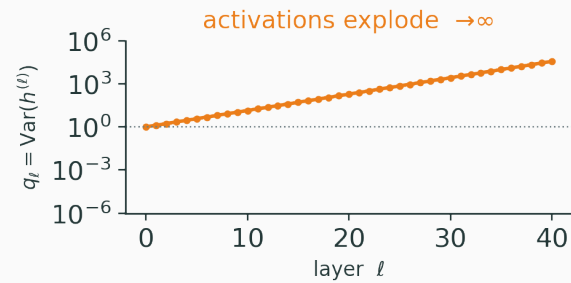
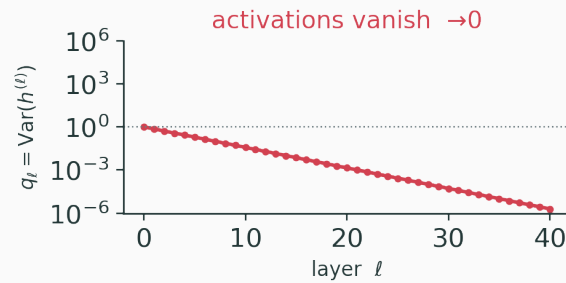
the problem is in the **forward and backward signals**, before optimization even begins

Push a signal through L layers and **track two numbers** per layer:

$$q_\ell = \text{Var}(h^{(\ell)}) \quad (\text{activation scale, forward}), \quad g_\ell = \|\partial \mathcal{L} / \partial h^{(\ell)}\|^2 \quad (\text{gradient scale, backward})$$

With naive init, **both** q_ℓ and g_ℓ drift **exponentially** with depth — either $\rightarrow 0$ (vanish) or $\rightarrow \infty$ (explode). A healthy net keeps them $\approx$ constant.

Four failure modes



activations and gradients can each **vanish** or **explode** — four ways depth breaks

Forward — each layer applies a linear map then a nonlinearity:

$$z^{(\ell)} = W_{\ell} h^{(\ell-1)}, \quad h^{(\ell)} = \varphi(z^{(\ell)}).$$

Backward — backprop pushes the adjoint through the transpose and the local slope:

$$\bar{h}^{(\ell-1)} = W_{\ell}^{\top} (\bar{h}^{(\ell)} \odot \varphi'(z^{(\ell)})).$$

$\bar{h}^{(\ell)} := \partial \mathcal{L} / \partial h^{(\ell)}$ — the same W and φ' control **both** directions

Compose the per-layer Jacobians $J_\ell = \partial h^{(\ell)} / \partial h^{(\ell-1)}$:

$$\frac{\partial h^{(L)}}{\partial x} = J_L J_{L-1} \cdots J_1.$$

A product of L matrices is governed by its **singular values** s :

$s < 1$ at each layer $\Rightarrow$ product $\rightarrow 0 \Rightarrow$
vanishing.

$s > 1$ at each layer $\Rightarrow$ product $\rightarrow \infty \Rightarrow$
exploding.

we need each layer's Jacobian to have singular values ≈ 1

Four interventions, one goal

Problem	Intervention
wrong starting scale of signals	initialization — set $\text{Var}(w)$
local slope kills the gradient	activations — shape the Jacobian φ'
scale drifts during training	normalization — re-standardize each layer
gradient path too long	residual connections — add identity shortcuts

all four exist to keep q_ℓ and g_ℓ near constant with depth

Activations and gradient flow

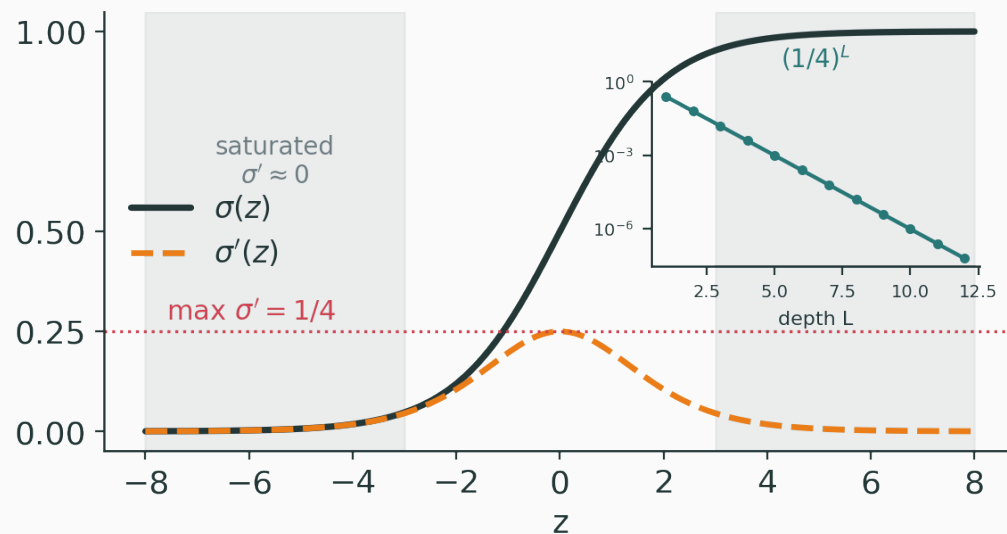
What an activation must do

A good activation φ has to be three things at once:

- **nonlinear** — else a stack of layers collapses to one linear map;
- **gradient-friendly** — φ' not tiny, so backprop survives depth;
- **cheap** — evaluated billions of times per step.

If φ' is small over most of its range, deep gradients **vanish** — the activation itself becomes the bottleneck.

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)).$$



the derivative peaks at $1/4$ and is ≈ 0 once $|z|$ is large — **saturation**

The peak slope is only $1/4$, so a chain of sigmoids can only **shrink** gradients:

$$\prod_{\ell=1}^L \sigma'(z_{\ell}) \leq \left(\frac{1}{4}\right)^L.$$

For $L = 10$: $(1/4)^{10} \approx 9.5 \times 10^{-7}$ — gradients are essentially gone.

Sigmoid is also **not zero-centred** (outputs in $(0, 1)$): activations are all-positive, biasing gradients and slowing learning.

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad \tanh'(z) = 1 - \tanh^2(z).$$

- **zero-centred** — outputs in $(-1, 1)$, fixing sigmoid's bias problem;
- max derivative is 1 (not $1/4$) — gradients survive a bit longer.

Still **saturates** for large $|z|$: $\tanh' \rightarrow 0$. Better than sigmoid, but deep tanh stacks still vanish.

$$\text{ReLU}(z) = \max(0, z), \quad \text{ReLU}'(z) = \mathbb{1}[z > 0] \in \{0, 1\}.$$

- **no positive saturation** — for $z > 0$ the slope is exactly 1, so gradients pass **unattenuated**;
- **sparse** — about half the units output 0;
- **cheap** — a single comparison.

the default hidden activation for MLPs and CNNs

If a unit's pre-activation is **always** negative, it is stuck:

$$z < 0 \Rightarrow h = 0, \quad \partial h / \partial z = 0 \quad (\text{no gradient, ever}).$$

Causes: too-large learning rate, bad init, or a large **negative bias** pushing z below 0.

A **dead** ReLU never recovers — its gradient is zero, so it is never updated. Watch the **fraction of zero activations** as a diagnostic.

Leaky ReLU and PReLU

Give the negative side a small slope so the unit can recover:

$$\text{LeakyReLU}(z) = \begin{cases} z & z > 0 \\ az & z \leq 0 \end{cases}, \quad a \approx 0.01.$$

- the negative branch has slope $a > 0 \Rightarrow$ **nonzero gradient** $\Rightarrow$ no permanent death;
- **PReLU** makes a a **learned** parameter per channel.

A cheap insurance policy against dead units, at the cost of one extra hyperparameter (or parameter).

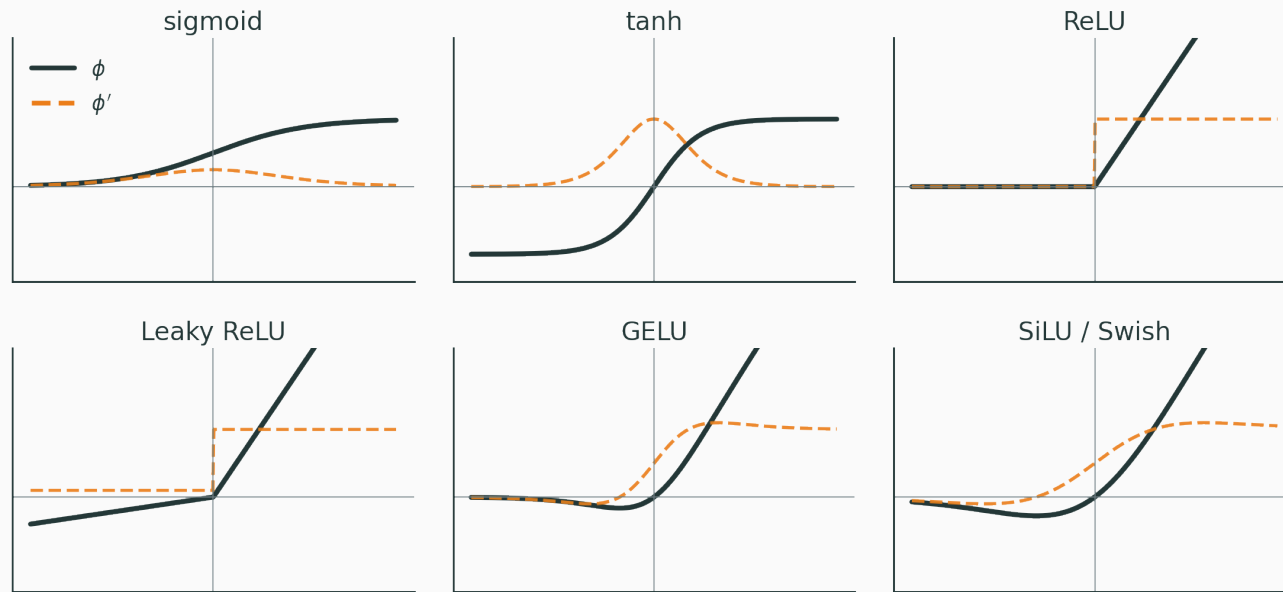
Gate the input by a smooth probability instead of a hard step:

$$\text{GELU}(z) = z \Phi(z), \quad \text{SiLU}(z) = z \sigma(z),$$

where Φ is the standard-normal CDF.

- smooth everywhere $\Rightarrow$ well-behaved second-order behaviour;
- **nonzero gradient** for small negative z (unlike ReLU);
- the default in **Transformers** (GELU) and many modern CNNs (SiLU / Swish).

The activation zoo



solid ϕ , dashed ϕ' — ReLU/Leaky/GELU/SiLU keep a healthy slope where sigmoid/tanh flatten

Activation	0-centred	Saturates	Neg. grad	Typical use
sigmoid	no	both sides	≈ 0	gates, binary output
tanh	yes	both sides	small	older RNN hidden
ReLU	no	no (pos.)	0 (dies)	CNN / MLP default
Leaky / PReLU	no	no	az (alive)	avoid dead units
GELU / SiLU	$\approx$	no	smooth	Transformers, modern CNNs

rule of thumb: ReLU/GELU by default; tanh/sigmoid only where you specifically want a bounded output

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Swap the activation and sweep depth; watch the gradient that reaches layer 0 collapse for sigmoid/tanh and survive for ReLU/GELU.

Compare the gradient-scale traces for a saturating activation and a modern activation.

Which pair most directly improves gradient flow in a deep sigmoid network?

A. More sigmoid layers and larger negative biases

C. Smaller batches and no initialization

B. A gradient-friendly activation and variance-preserving initialization

D. Remove all nonlinearities

Correct: B — A gradient-friendly activation and variance-preserving initialization

Why: ReLU/GELU-like slopes avoid widespread saturation, and Xavier/He-style scaling prevents signal scale from compounding badly at the start.

Initialization: the starting scale

Why not initialize to all zeros?

Set every weight equal (e.g. all zero). Two units in a layer see the **same** input and the same weights, so:

same activation $\Rightarrow$ same gradient $\Rightarrow$ same update (forever).

Symmetry is never broken — every unit in a layer stays identical, so the layer has the expressive power of **one** unit.

Random weights break the symmetry. **Biases** can safely start at 0 — the random weights already differentiate the units.

Study one pre-activation with n zero-mean, independent inputs and weights:

$$z = \sum_{i=1}^n w_i x_i.$$

With $\mathbb{E}[w_i] = 0$ and $w \perp x$:

$$\mathbb{E}[z] = \sum_{i=1}^n \mathbb{E}[w_i] \mathbb{E}[x_i] = 0.$$

zero-mean weights $\Rightarrow$ zero-mean pre-activations – now control the variance

Variance of a sum of independent, zero-mean terms:

$$\text{Var}(z) = \sum_{i=1}^n \text{Var}(w_i x_i) = \sum_{i=1}^n \text{Var}(w) \text{Var}(x).$$

All n terms are identical, so

$$\text{Var}(z) = n \text{Var}(w) \text{Var}(x).$$

$$\text{Var}(z) = n_{\text{in}} \text{Var}(w) \text{Var}(x)$$

To keep the signal scale **unchanged** across a layer we need $\text{Var}(z) = \text{Var}(x)$:

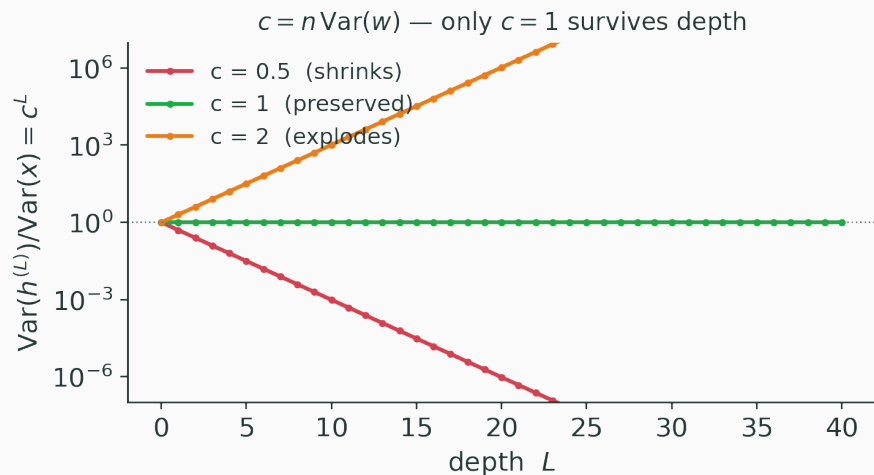
$$n_{\text{in}} \text{Var}(w) \approx 1 \quad \Rightarrow \quad \text{Var}(w) \approx \frac{1}{n_{\text{in}}}.$$

Scale the weights by the **fan-in**. Too big amplifies each layer; too small attenuates each layer.

The wrong scale compounds

Write $c = n_{\text{in}} \text{Var}(w)$. Across L layers the variance is **multiplied** L times:

$$\text{Var}(h^{(L)}) \approx c^L \text{Var}(x).$$



$$c = 0.5, L = 20 : 0.5^{20} \approx 9.5 \times 10^{-7}$$

$$c = 2, L = 20 : 2^{20} \approx 1.0 \times 10^6$$

only $c = 1$ survives depth; a small miss is fatal after 20 layers

Backprop multiplies by $W^\top$, so the **gradient** variance follows the same law with fan-**out**:

$$\text{Var}(\bar{h}^{(\ell-1)}) \approx n_{\text{out}} \text{Var}(w) \text{Var}(\bar{h}^{(\ell)}).$$

Preserving the **backward** signal wants

$$\text{Var}(w) \approx \frac{1}{n_{\text{out}}}.$$

Forward wants $1/n_{\text{in}}$, backward wants $1/n_{\text{out}}$ — and usually $n_{\text{in}} \neq n_{\text{out}}$. We cannot satisfy both exactly.

Compromise between the two constraints — take the **harmonic-style** average:

$$\text{Var}(w) = \frac{2}{n_{\text{in}} + n_{\text{out}}}.$$

Two equivalent samplers:

$$w \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}} + n_{\text{out}}}\right), \quad w \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right).$$

Xavier balances forward and backward variance — designed for tanh / linear layers

Xavier assumes φ is linear near 0. ReLU **zeros out half** the inputs:

$$h = \text{ReLU}(z) \Rightarrow \mathbb{E}[h^2] \approx \frac{1}{2}\mathbb{E}[z^2].$$

So a ReLU layer **halves** the signal variance — Xavier's c becomes ≈ 0.5 , and the signal decays.

We must put the lost factor of 2 **back** into the weight variance.

Demand variance preservation **through** the ReLU:

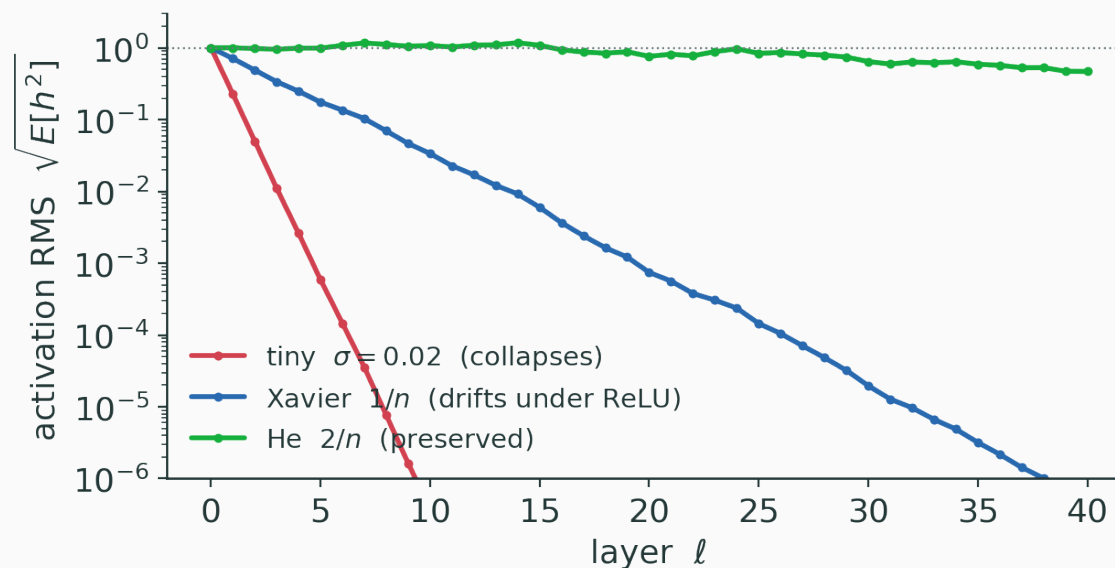
$$\frac{1}{2} n_{\text{in}} \text{Var}(w) \approx 1 \quad \Rightarrow \quad \text{Var}(w) = \frac{2}{n_{\text{in}}}, \quad \text{Std}(w) = \sqrt{\frac{2}{n_{\text{in}}}}.$$

He init: $\text{Std}(w) = \sqrt{2/n_{\text{in}}}$ — the default for ReLU networks

Generalized for a leaky slope a :

$$\text{Var}(w) = \frac{2}{(1 + a^2) n_{\text{in}}}.$$

Signal flow under three inits



He holds the activation scale flat through a deep ReLU net; a tiny σ collapses; Xavier drifts down

Layer with $n_{\text{in}} = 100$, unit-variance inputs, so $\text{Var}(z) = 100 \text{Var}(w)$. Case (A), too small

Init example: the two failures

$\text{Std}(w) = 0.01$, so $\text{Var}(w) = 10^{-4}$:

$\text{Var}(z) = 100 \cdot 10^{-4} = 0.01$ (shrinks $100 \times$)

Case (B), too large

Init example: the two failures

$\text{Std}(w) = 1$, so $\text{Var}(w) = 1$:

$\text{Var}(z) = 100 \cdot 1 = 100$ (explodes $100 \times$)

off by $100 \times$ in **one** layer; over 20 layers that compounds to $100^{\pm 20}$ – hopeless.

Same layer ($n_{\text{in}} = 100$, $\text{Var}(z) = 100 \text{Var}(w)$) — now compare all four schemes:

Scheme	$\text{Var}(w)$	$\text{Var}(z)$	Effect
(A) Std = 0.01	10^{-4}	0.01	shrinks $100 \times$ — vanishes
(B) Std = 1	1	100	explodes $100 \times$
(C) Xavier 1/100	0.01	1	preserved (linear)
(D) He 2/100	0.02	2	pre-ReLU 2, post-ReLU $\mathbb{E}[h^2] \approx 1$

(A)/(B) are off by $100 \times$ in one layer; (C)/(D) preserve the scale — all verified numerically

For a conv layer the fan-in counts **every input that feeds one output**:

$$n_{\text{in}} = k_h \cdot k_w \cdot C_{\text{in}}.$$

A 3×3 kernel over 64 input channels:

$$n_{\text{in}} = 3 \cdot 3 \cdot 64 = 576, \quad \text{Std}_{\text{He}} = \sqrt{\frac{2}{576}} \approx 0.0589.$$

same rule, correct fan-in — the constant just changes

Initialization is not a guarantee

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Good init makes training **start** in a healthy regime — but only at $t = 0$.

The derivations assume zero-mean, independent, unit-scale inputs. During training weights **shift**, correlations build, and the guarantees **erode**.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep the init scale and watch the per-layer activation std light up green (healthy) or red (vanishing / exploding) across a deep stack.

**Normalization: scale during
training**

Initialization sets the signal scale at $t = 0$; then optimization moves the weights and the guarantee erodes. We need to **re-standardize** activations at **every** step. For a batch of a quantity x , subtract the mean and divide by the std:

$$\mu = \frac{1}{B} \sum_{i=1}^B x_i, \quad \sigma^2 = \frac{1}{B} \sum_{i=1}^B (x_i - \mu)^2, \quad \hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \varepsilon}}.$$

$\hat{x}$ has mean 0 and variance 1 — a controlled scale, by construction

$\varepsilon \approx 10^{-5}$ guards against dividing by zero

Forcing mean 0, variance 1 can be **too** restrictive. Add a learnable affine map back:

$$y = \gamma \hat{x} + \beta.$$

- γ rescales, β re-shifts — both **learned**;
- the network can **recover the identity** ($\gamma = \sqrt{\sigma^2 + \varepsilon}$, $\beta = \mu$) if normalizing hurt.

normalize for a stable scale, then let the model choose the scale it wants

Which axis do we normalize over?

Given a batch of activations $X \in \mathbb{R}^{B \times D}$ (B examples, D features), we can normalize:

- **down each feature** (over the batch) → **BatchNorm**;
- **across each example** (over the features) → **LayerNorm**.

The **axis you reduce over** is the entire difference between the normalization schemes.

Normalize **each feature** j using statistics **over the batch**:

$$\mu_j = \frac{1}{B} \sum_{i=1}^B x_{ij}, \quad \sigma_j^2 = \frac{1}{B} \sum_{i=1}^B (x_{ij} - \mu_j)^2, \quad y_{ij} = \gamma_j \hat{x}_{ij} + \beta_j.$$

One $(\mu_j, \sigma_j, \gamma_j, \beta_j)$ **per feature**. Each column of the batch matrix is standardized independently.

BatchNorm normalizes down each feature



orange column = one feature j ,
normalized using its mean/variance
over the whole batch.

rows = examples in the batch (B)

columns = features (D)

reduce **down** each column

Treat every spatial location as another sample. Normalize **per channel** over (B, H, W) :

μ_c, σ_c^2 computed over all $B \cdot H \cdot W$ activations of channel c .

One statistic **per channel**, shared across space — this is why BatchNorm is so natural in CNNs.

At **training** time BatchNorm uses the **batch** statistics. At **inference** a single example has no batch — so keep a **running average**:

$$\mu_{\text{run}} \leftarrow \rho \mu_{\text{run}} + (1 - \rho) \mu_B, \quad \sigma_{\text{run}}^2 \leftarrow \rho \sigma_{\text{run}}^2 + (1 - \rho) \sigma_B^2.$$

`model.train()` uses batch stats; `model.eval()` uses the frozen running stats.
Forgetting to switch is a classic bug.

Feature values $x = [1, 2, 3, 4]$ across a batch of 4, learnable $\gamma = 2, \beta = 1$. Mean — average over the batch:

$$\mu = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

Variance — average squared deviation from μ :

$$\sigma^2 = \frac{1}{4}(2.25 + 0.25 + 0.25 + 2.25) = 1.25$$

next: standardize with these stats, then apply the learnable γ, β .

Standardize each value with $\mu = 2.5$, $\sigma^2 = 1.25$ (so $\sqrt{\sigma^2} \approx 1.118$):

$$\hat{x} = \frac{x - 2.5}{\sqrt{1.25}} \approx [-1.342, -0.447, 0.447, 1.342].$$

Apply the learnable affine $y = \gamma\hat{x} + \beta = 2\hat{x} + 1$:

$$y \approx [-1.683, 0.106, 1.894, 3.683].$$

$\hat{x}$ is mean-0 / var-1; γ, β then place it where the network wants – verified numerically

Why does BatchNorm help?

The original story was “**reducing internal covariate shift**” — the layer-input distribution moving during training.

Later work found that explanation **does not hold up**: the better-supported reason is that BN makes the optimization **landscape smoother** (well-conditioned), so larger, more stable steps are possible.

Teach it as **smoother optimization + a controlled scale**, not as “fixing covariate shift”.

BatchNorm's limitations

BatchNorm couples every example in the batch:

- **batch-dependent** — an example's output depends on **who else** is in the batch;
- **small-batch noise** — statistics are unreliable for tiny batches;
- **train $\neq$ inference** — needs running stats and a mode switch;
- **awkward for sequences** — variable lengths, per-step statistics.

these are exactly the cases where LayerNorm wins

Normalize **each example over its own features** — no batch involved:

$$\mu_i = \frac{1}{D} \sum_{j=1}^D x_{ij}, \quad \sigma_i^2 = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \mu_i)^2, \quad y_{ij} = \gamma_j \hat{x}_{ij} + \beta_j.$$

- **batch-independent** — identical result for batch size 1 or 1000;
- **train = inference** — no running statistics needed.

LayerNorm normalizes across each example



orange row = one example i ,
normalized using the mean/variance **of
its own features**.

reduce **across** each row — the batch axis is never
touched

Aspect	BatchNorm	LayerNorm
normalize over	batch (per feature)	features (per example)
batch-dependent?	yes	no
train = inference?	no (running stats)	yes
small batch	noisy / unstable	unaffected
sequences	awkward	natural
typical use	CNNs	Transformers, RNNs

Drop the mean-centring; rescale by the root-mean-square only:

$$\text{RMS}(x) = \sqrt{\frac{1}{D} \sum_{j=1}^D x_j^2}, \quad y = \gamma \frac{x}{\text{RMS}(x)}.$$

Cheaper than LayerNorm (no mean, no subtraction), and works about as well — common in recent large language models.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/norm-comparison

Toggle Batch / Layer / RMS norm on the same $B \times D$ tensor and watch which axis gets reduced — then compare train-time vs eval-time behaviour for BatchNorm.

Compare normalizing one example's features with normalizing each feature across a batch.

Residual connections: short paths

He et al. (2015): a **56-layer** plain net had **higher training error** than a **20-layer** one.

Not overfitting — the **training** error is worse, not just the test error.

It is an **optimization** failure: the deeper net **cannot even fit** the data the shallow one fits.

deeper should be at least as easy — yet plain stacks get **harder to optimize**

A deep block should be able to represent the **identity** easily. Reparameterize the target $H(x)$:

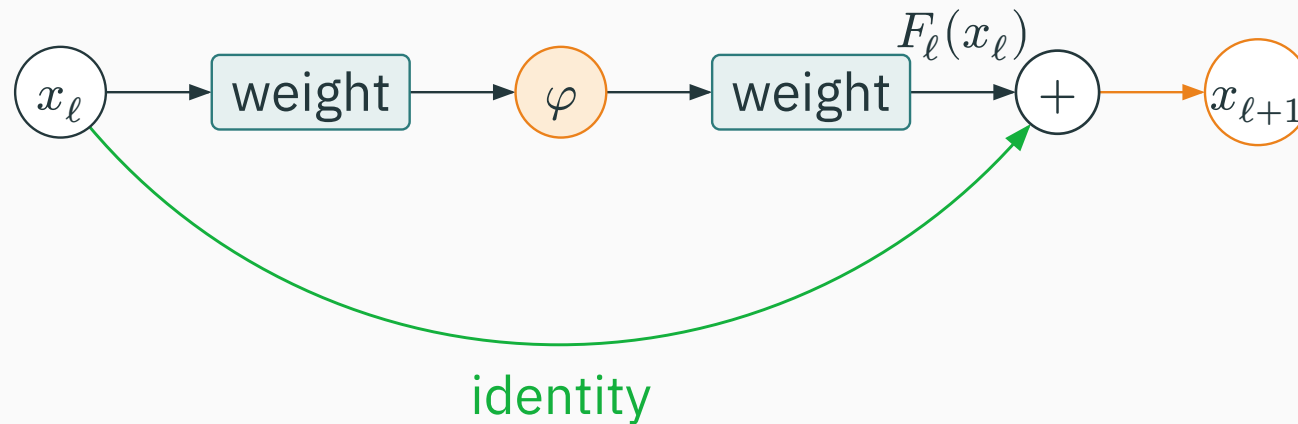
$$F(x) := H(x) - x \quad \Rightarrow \quad H(x) = x + F(x).$$

Instead of learning H from scratch, the block learns the **residual** F — the **change** to apply.

If the best thing to do is “nothing”, the block only needs $F \approx 0$ — far easier than learning $H \approx \text{identity}$ from random weights.

Stack of blocks, each adding its output to its input:

$$x_{\ell+1} = x_{\ell} + F_{\ell}(x_{\ell}).$$



the **green identity shortcut** carries x_{ℓ} straight to the addition, skipping the branch

Unroll the recurrence from layer ℓ to layer L :

$$x_L = x_\ell + \sum_{i=\ell}^{L-1} F_i(x_i).$$

every layer has a **direct, additive contribution to the output**

the input reaches the top through the identity path, never multiplied away

Differentiate $x_{\ell+1} = x_{\ell} + F_{\ell}(x_{\ell})$:

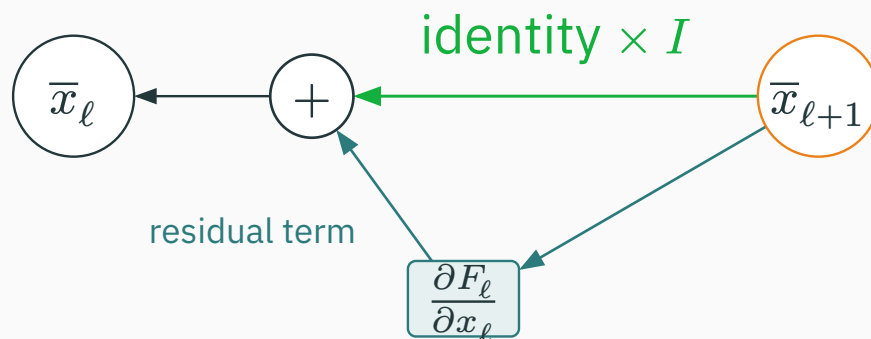
$$\frac{\partial \mathcal{L}}{\partial x_{\ell}} = \frac{\partial \mathcal{L}}{\partial x_{\ell+1}} \left(I + \frac{\partial F_{\ell}}{\partial x_{\ell}} \right) = \underbrace{\frac{\partial \mathcal{L}}{\partial x_{\ell+1}}}_{\text{identity term}} + \underbrace{\frac{\partial \mathcal{L}}{\partial x_{\ell+1}} \frac{\partial F_{\ell}}{\partial x_{\ell}}}_{\text{residual term}}.$$

the identity term passes the gradient through with **no matrix multiply**

even if the branch Jacobian vanishes, the I keeps a clean path back

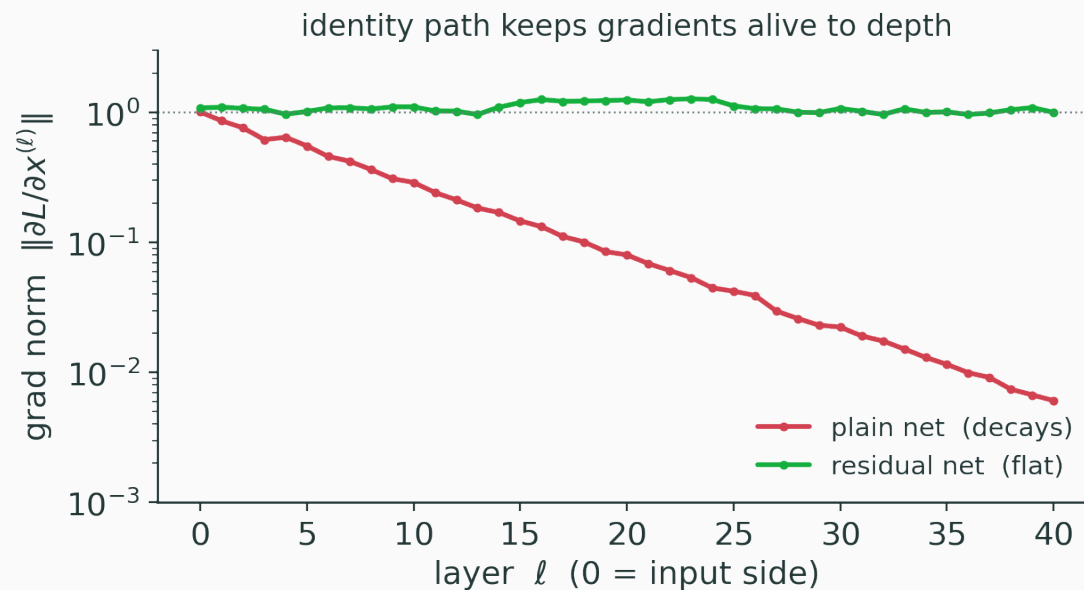
The gradient's two paths

Backward, the upstream gradient $\bar{x}_{\ell+1}$ splits and **adds** at the input:



the **green identity path** delivers $\bar{x}_{\ell+1}$ untouched — so $\bar{x}_\ell$ stays alive even if the **residual term** shrinks

Residuals keep gradients alive



plain net: gradient decays with depth · residual net: the identity path holds it flat

Take a scalar block $y = x + wx^2$ with the shortcut. Differentiate:

$$\frac{dy}{dx} = 1 + 2wx.$$

At $w = 0.1, x = 2$: $1 + 2(0.1)(2) = 1.4$. **Without** the shortcut ($y = wx^2$): $\frac{dy}{dx} = 2wx = 0.4$.

the identity keeps a 1 in the gradient – it cannot vanish below that

When F_ℓ changes width or resolution, the identity no longer fits. Project it:

$$x_{\ell+1} = P_\ell x_\ell + F_\ell(x_\ell).$$

- P_ℓ is a learned **linear projection** (a 1×1 conv in CNNs, or a stride to downsample);
- used only at the blocks where dimensions change; elsewhere $P_\ell = I$.

Where does the activation (and norm) sit relative to the add?

Post-activation (original ResNet):

$$x_{\ell+1} = \varphi(x_{\ell} + F_{\ell}(x_{\ell})).$$

Pre-activation: norm and φ go **inside** F_{ℓ} , keeping the shortcut a **clean** identity.

pre-activation keeps the skip path unobstructed — better for very deep nets

Residuals are not magic

A shortcut alone does not fix everything:

- the branch still needs sensible **scale and init** (start $F \approx 0$);
- **norm placement** around the block matters (pre- vs post-);
- the **learning rate** still interacts with everything.

Fixup init trains deep ResNets **without** normalization — by carefully scaling the residual branches. Init, norm and residuals are three **coupled** knobs, not independent fixes.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/resnet

Toggle the skip connections on and off in a deep net and watch the backward gradient norm go from decaying (plain) to flat (residual), layer by layer.

Trace the identity and residual contributions separately as the gradient moves backward.

In a residual block, which term supplies the direct gradient path?

A. The identity term $\partial \frac{\mathcal{L}}{\partial} x_{\ell+1}$

B. Only the residual Jacobian $\partial \frac{F_{\ell}}{\partial} x_{\ell}$

C. The learning-rate multiplier

D. The batch mean

Correct: A – The identity term

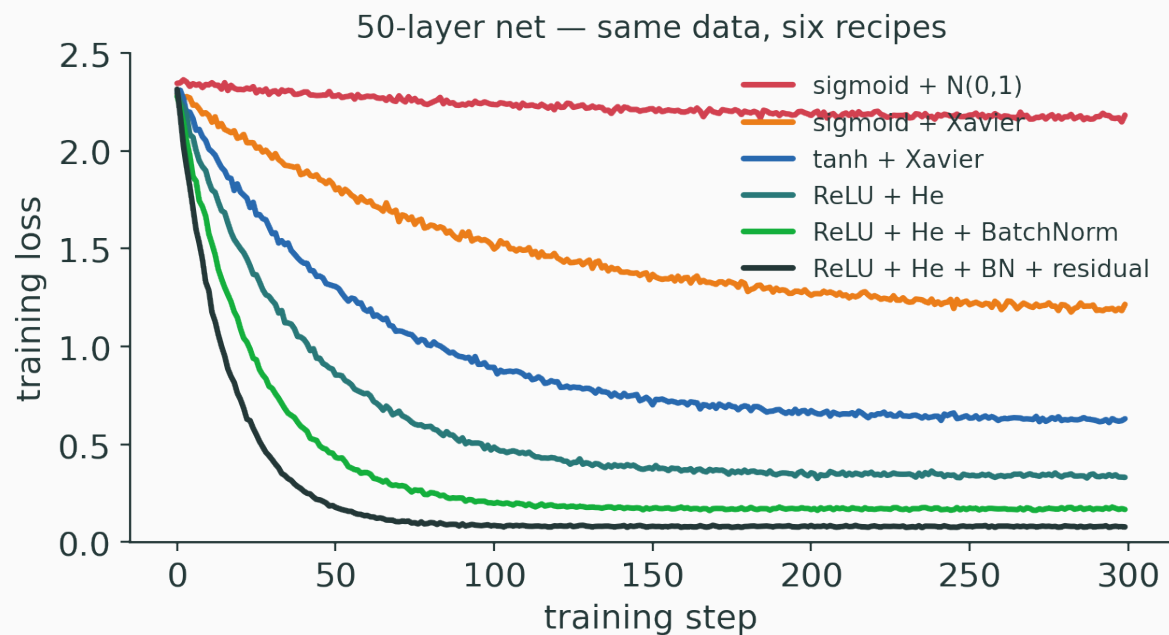
Why: Differentiating $x_{\ell+1} = x_{\ell} + F_{\ell}(x_{\ell})$ gives an additive identity contribution, so the branch need not carry the whole gradient.

Putting it together

Same 50-layer net and data, six recipes stacked from naive to modern:

Activation	Init	Extras	Outcome
sigmoid	$\mathcal{N}(0, 1)$	—	dead — no learning
sigmoid	Xavier	—	learns, very slow
tanh	Xavier	—	trains, still slow
ReLU	He	—	trains reliably
ReLU	He	BatchNorm	faster, stable
ReLU	He	BN + residual	fastest, deepest

each intervention removes a different obstacle to optimization



naive sigmoid never moves; each added technique lowers and speeds the loss

What each technique changes

Technique	What it controls
initialization	the starting distribution of signals
activation	the local nonlinearity and its derivative φ'
normalization	the scale and parameterization during training
residual	the path length for signal and gradient

These knobs are coupled

They are **not** independent tricks:

- ReLU $\Rightarrow$ use **He**, not Xavier (the factor of 2);
- BatchNorm **reduces** but does not **remove** init sensitivity;
- residual branches want **smaller** init (start near 0);
- LayerNorm **placement** changes the residual stream;
- the **learning rate** interacts with all of the above.

Change one and you often have to retune another — treat them as a **system**, not a checklist.

When a deep net will not train, **log per layer** and look for the outlier:

```
for name, h in activations.items():
    log(name, h.mean(), h.std(), h.min(), h.max())    # signal scale
    log(name, (h == 0).float().mean())              # % dead (ReLU)
for name, p in model.named_parameters():
    log(name, p.grad.norm() / (p.norm() + 1e-8))    # grad-to-weight ratio
    log(name, (~torch.isfinite(p.grad)).any())      # NaN / Inf guard
```

healthy: std $\approx$ const with depth, grad/weight ratio $\approx 10^{-2}$ to 10^{-3} everywhere, no NaNs

Symptom	Likely issue	Fix
activation std $\rightarrow 0$ with depth	init too small / saturation	He init, ReLU/GELU
activation std $\rightarrow \infty$ / NaN	init too large / no norm	scale down, clip, add norm
high % zero activations	dead ReLU (LR / neg bias)	lower LR, LeakyReLU
tiny grad in early layers	vanishing gradient	residual, norm, ReLU
loss diverges in first steps	LR too high / no warmup	warmup, lower LR, clip

Setting	Sensible default recipe
MLP	ReLU / GELU + He init + (optional LayerNorm)
CNN	He init + ReLU / SiLU + BatchNorm + residual
Transformer	LayerNorm / RMSNorm + residual + GELU / SiLU + warmup

start here, then adjust — these are starting points, not laws

What **not** to overclaim

- BatchNorm does **not** “solve internal covariate shift” — it smooths optimization;
- residuals do **not** prevent **all** vanishing — they add a path, not a guarantee;
- ReLU **can** effectively **die** — watch the zero fraction;
- He does **not** preserve variance **exactly** — it is an approximation that erodes;
- normalization is **not always** required (Fixup, careful init);
- **deeper is not always better** — depth has to be earned.

Say the fix out loud before you read it:

Cue	Recall
tanh / sigmoid	saturation $\Rightarrow$ vanishing gradients
ReLU on the negative side	zero gradient $\Rightarrow$ can die
$\text{Var}(W) = 1/n_{\text{in}}$	Xavier (linear / tanh)
$\text{Var}(W) = 2/n_{\text{in}}$	He (ReLU)
$x + F(x)$	residual block $\Rightarrow$ identity gradient path

Final mental model — four moves

initialization = start in a trainable regime

activations = preserve a nonlinear gradient

normalization = stabilize scale + optimization

residuals = shorten the signal / gradient paths

all four keep signals alive forward and gradients alive backward

Now the network can be **optimized**.

Next: keeping it from fitting the training data too well — **regularization
and generalization**.